

Aakarsh Sangar

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Engineering Portfolio: aakarshsangar.com

EDUCATION

Purdue University

West Lafayette, IN | Class of 2030

First-Year Engineering; Intended B.S. Mechanical Engineering

EXPERIENCES

ModalAI

San Diego, CA | Sep 2024 - Jan 2025

Drone Hardware Intern

- One of three hardware interns on a 5-month pre-release Stinger VisionM FPV kit, designing sensor-interface and power-distribution updates for an unreleased commercial drone.
- Integrated onboard sensors and embedded processors, then tested drone kits through bring-up and reconfiguration to support live perception and control workflows.
- Designed and 3D printed missing drone components, resolving prototype fit gaps and improving kit assembly/user setup before customer-facing documentation.
- Authored one assembly and bring-up manual, documenting mechanical, hardware, and software prototype iterations so early users could repeat the setup process.

TECHNICAL PROJECTS

EPICS Hippotherapy Machine

West Lafayette, IN | Sep 2026 - Present

Mechanical Design Project

- Contributing to a robotic horseback-motion simulator for Therapeion Therapeutic Riding Center, using mechanism design to support core and trunk therapy.
- Applying CAD analysis to reproduce rhythmic forward-rocking horse motion, translating therapy requirements into linkage and frame design constraints.
- Developing Fusion Simulation FEA studies for rider-load cases to reduce deformation or failure risk before children use the mechanical simulator.

Planetary-Drive Robotic Arm

San Diego, CA | Jun 15 - Aug 13, 2026

Onshape, ROS 2

- Created a multi-joint robotic arm around a custom planetary-drive actuator concept, estimating worst-case joint loads to guide actuator placement, gearbox sizing, and arm geometry.
- Modeled four independently actuated base, link, wrist, and gripper joints in Onshape to evaluate axis placement, joint limits, and packaging.
- Built a ROS 2/URDF simulation path from CAD geometry to validate joint frames, mesh alignment, and motion interfaces before physical iteration.

Actuator Study & Simulated Arm

University of California San Diego | May 31 - Jul 25, 2025

UCSD faculty-guided | Onshape, URDF, PyBullet

- Compared five actuator classes across torque density, controllability, power, and integration to choose better-fit mechanisms for arm joints.
- Exported an articulated Onshape arm to a URDF tree, converting CAD assemblies into simulation-ready links, joints, and mesh references.
- Validated axes, limits, and mesh frames in PyBullet so simulated motion matched the intended mechanical joint structure.

FIRST Tech Challenge 18263 HIGH VOLTAGE

San Diego, CA | May 2020 - May 2026

CAD Lead, Build Team

- Led Onshape CAD and mechanical iteration across six FTC seasons, building a multidirectional chassis that supported PID-stabilized driver control.
- Integrated kinematics-based turret aiming into the robot architecture, improving repeatable target alignment through mechanical packaging and geometry.
- Iterated intake, transfer, turret, and drivetrain packaging with build teammates, contributing to 2x Design, Motivate, Innovate, and Inspire 3rd Place awards.

Faultline

University of California Berkeley | Jun 20 - Jun 21, 2026

CalHacks AI

- Built a research assistant to compare conflicting claims across academic papers, structuring source evidence so users could evaluate disagreements more quickly.

Purdue Electric Racing

West Lafayette, IN | Sep 2026 - Present

Drivetrain Team

- Joining drivetrain design to apply mechanical CAD in Siemens NX on upright, gearbox, and packaging concepts for a Formula-style EV.
- Preparing Ansys FEA studies on drivetrain and upright geometry to check stiffness, deformation, and structural margins before fabrication.
- Evaluating torque-speed tradeoffs and GD&T constraints so suspension-to-wheel-hub interfaces can be designed for manufacturability and assembly.

TECHNICAL SKILLS

CAD & Analysis: Onshape Certified, Siemens NX, Fusion Simulation, Ansys FEA, GD&T, tolerance-aware design

Mechanical Systems: Actuator/transmission selection, planetary drives, gearbox concepts, upright design, structural integrity analysis

Programming for Engineering: Python, C/C++, Java, Arduino, Matplotlib, URDF, PyBullet